

PROJECT PRESENTATION

Hotel Booking & Management System

A Java Swing desktop application for managing hotel rooms and customer bookings

ROOMS

Room information

BOOKING

Customer bookings

SEARCH

Room lookup

DISPLAY

Status & sorting

Introduction & Problem Statement

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Why the system is needed

PROBLEM STATEMENT

- Manual room booking can cause availability errors.
- Customer booking information must be tracked consistently.
- Room-number searching should be quick and reliable.
- Cancellation must update both booking and room status.

PROPOSED SOLUTION

- A simple Java Swing desktop application.
- Room information maintained using an array.
- HashMap provides room-number-based searching.
- LinkedList stores booking records.
- Cancellation releases the room for future booking.

CORE OPERATIONS

Book Room

Cancel Booking

Available Rooms

All Rooms

Data Structures

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How each collection supports a specific requirement

ARRAY — ROOM STORAGE

- Stores predefined Room objects.
- Contains room number, type, price and availability.
- Used to scan rooms for availability.

LINKEDLIST — BOOKINGS

- Stores Booking objects.
- Bookings can be added dynamically.
- Bookings can be removed during cancellation.

HASHMAP — ROOM SEARCH

- Integer room number = key.
- Room object = value.
- `get(201)` returns the Room 201 object.

TREEMAP — SORTED DISPLAY

- Room number is the key.
- Keys remain sorted for display.

GUI Components

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Implementation technologies and interface elements

SWING COMPONENTS

- JFrame — main application window.
- JPanel — groups interface sections.
- JTextField — accepts booking information.
- JButton — triggers system operations.
- JTextArea — displays system output.
- JScrollPane — makes output scrollable.
- JOptionPane — validation and error messages.

HOTEL BOOKING SYSTEM

Room Booking & Management

Booking Details

Customer Name:

Phone Number:

Room Number:

Number of Days:

Book Room

Cancel Booking

Available Rooms

All Rooms

System Output

Application Interface

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Main window of the Java Swing application

INTERFACE ELEMENTS

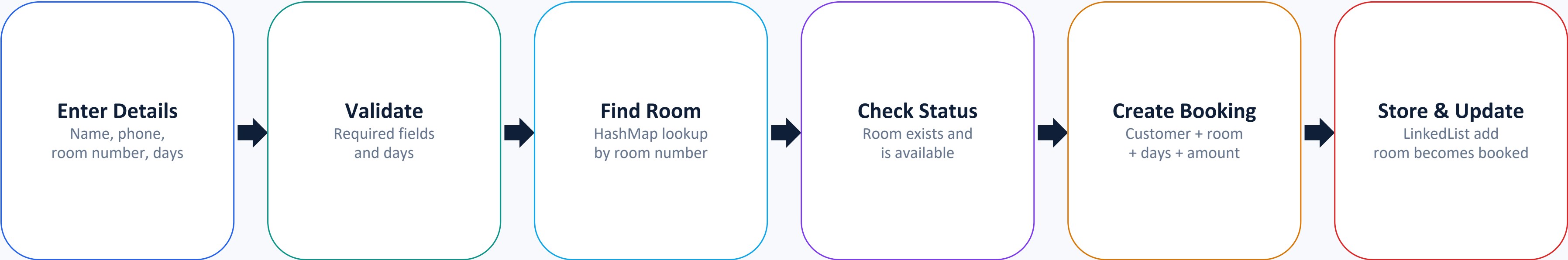
- Input fields collect customer and booking data.
- Buttons trigger booking, cancellation and display functions.
- Output area presents the result.

USER INTERACTION

- Book Room — create a booking.
- Cancel Booking — remove a booking.
- Available Rooms — show free rooms.
- All Rooms — show every room and status.

Room Booking: Program Flow

Validation occurs before a booking is stored

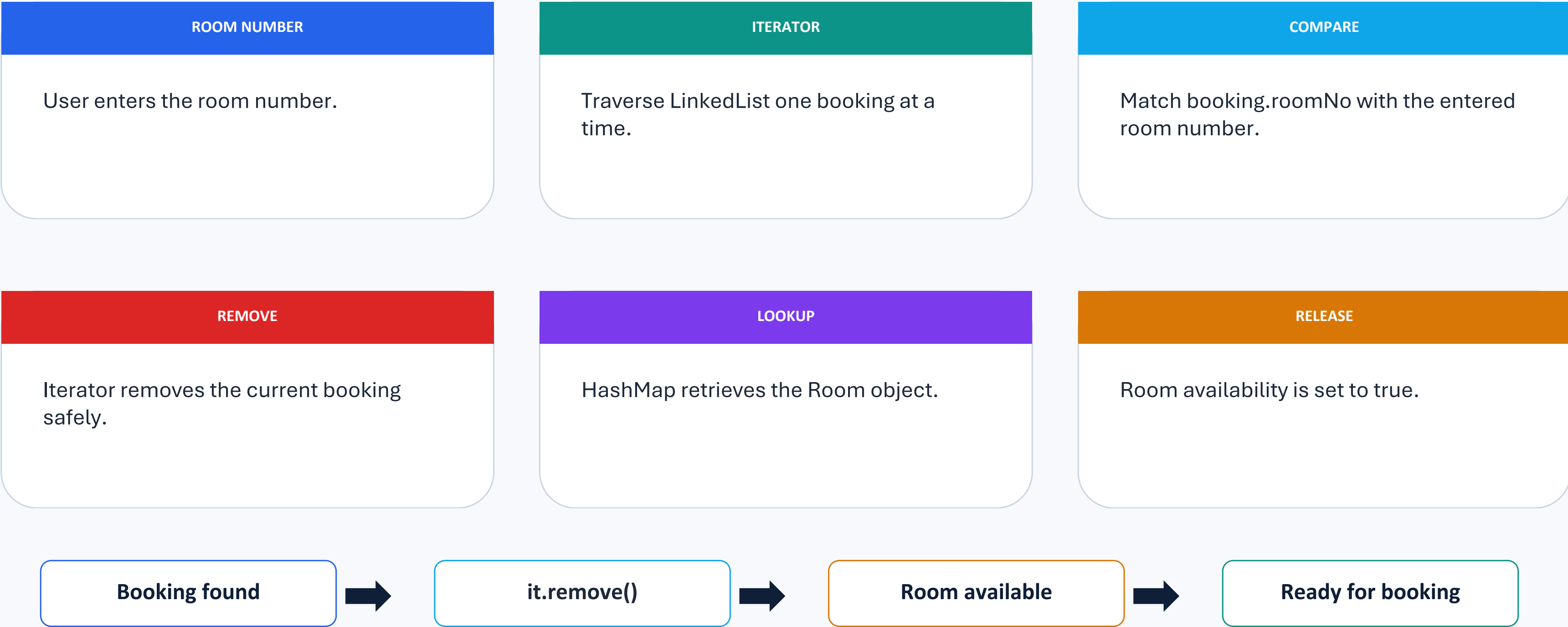


BILL CALCULATION

Total Bill = Room Price per Day × Number of Days

Booking Cancellation

Iterator-based traversal removes the matching booking



Display & Main

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Available Rooms, All Rooms and Clear Fields

AVAILABLE ROOMS

- Shows only rooms whose availability is true.
- After booking Room 201, it is no longer available.
- After cancellation, Room 201 becomes available again.

ALL ROOMS

- Uses TreeMap ordering by room number.
- Displays room type, price and Available/Booked status.

CLEAR FIELD

- Resets all input fields after a successful booking or cancellation.
- Clears customer name, phone number, room number, and number of days so the form is ready for the next operation.

Conclusion

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A complete Java Swing solution for hotel room and booking management

KEY OUTCOMES

- Room information is organized using an array.
- HashMap enables room-number-based lookup.
- LinkedList manages booking records dynamically.
- Iterator supports safe cancellation.
- TreeMap provides sorted room display.

SYSTEM WORKFLOW

- Customer enters booking information.
- Input is validated and the room is located.
- Bill is calculated and the booking is stored.
- Cancellation removes the booking and releases the room.
- The GUI reflects the current room status.

THANK YOU • QUESTIONS ARE WELCOME